Notes: Egan, Hortaçsu & Matvos (AER, 2017)

Deposit Competition and Financial Fragility: Evidence from the US Banking Sector

Contents

1 Big picture

- **Question.** How fragile is the US banking sector when uninsured depositors are run-prone, banks are differentiated, banks compete for deposits, and banks can endogenously default?
- **Central mechanism.** A bank's financial distress lowers demand for its uninsured deposits. The deposit outflow lowers bank profitability and raises default probability. If this feedback is strong enough, the same fundamentals can support multiple equilibria.
- **Empirical starting point.** Uninsured deposit demand declines with bank distress, while insured deposit demand does not. This asymmetry is essential because insured depositors are protected by FDIC insurance while uninsured depositors are exposed to loss or disruption in default.
- **Structural model.** Differentiated banks compete for insured and uninsured deposits by setting interest rates. Depositors choose banks in a logit demand system. Banks earn stochastic returns on deposits, have long-term debt, and choose whether to default.
- **Calibration.** The model is estimated and calibrated using a new dataset for 16 large US banks from 2002–2013, combining FDIC deposit quantities, RateWatch deposit rates, CDS-based default probabilities, branch/ATM/service quality controls, and bank balance sheet information.
- **Main quantitative finding.** The calibrated model has multiple equilibria with bank-run features. In March 2008, the observed equilibrium had Wachovia's default probability around 3.3 percent, but the same fundamentals also supported an equilibrium with Wachovia default probability around 52 percent.
- **Systemic finding.** Instability of one bank can spill over to other banks through deposit-rate competition, even without direct balance-sheet linkages. However, not all major banks become unstable at once because depositors value banking services and move toward relatively safer banks.
- **Policy finding.** Capital requirements below roughly 15–18 percent can permit very bad equilibria with large welfare losses. Higher requirements reduce the worst-case instability, but very high requirements can lower welfare by reducing valuable banking activity and consumer surplus.
- **Economic takeaway.** Deposit competition can both transmit distress and contain it. The same differentiated-product banking market that permits self-fulfilling runs also prevents complete collapse because some banks become relatively safer substitutes.

2 Incremental contribution of the paper

- **Quantitative bank-run contribution.** The paper moves from qualitative bank-run theory to a calibrated model of the largest US banks. Diamond-Dybvig-style models show that runs are possible; this paper asks whether US banking fundamentals and deposit elasticities make bad equilibria quantitatively relevant.
- **Demand-estimation contribution.** The paper estimates separate demand systems for insured and uninsured deposits and shows that uninsured deposits are sensitive to bank distress while insured deposits are not. This is a direct empirical measurement of run-prone deposit demand.
- **Industrial-organization contribution.** By embedding differentiated-product logit demand into a bank fragility model, the paper shows how product differentiation and deposit competition discipline the strength and propagation of runs.
- **Structural supply-side contribution.** The paper uses banks' first-order conditions and observed equilibrium behavior to calibrate unobserved deposit return means, deposit return volatilities, and costs of servicing insured deposits.
- **Multiple-equilibrium policy contribution.** Instead of evaluating policy only in the observed equilibrium, the paper evaluates the full set of equilibria supported by the same fundamentals. This is crucial because regulation may eliminate or create bad equilibria.

- **Regulatory contribution.** The model can compare FDIC insurance expansion, rate caps, capital requirements, too-big-to-fail expectations, costly capital adjustment, risk-free capital requirements, and run-prone insured depositor extensions in a common framework.
- **Welfare contribution.** The paper distinguishes banking stability from welfare. A policy can reduce default probabilities while lowering consumer surplus or valuable intermediation, so stability alone is not a sufficient welfare criterion.

3 Data and measurement

3.1 Bank sample and time period

The empirical bank-level dataset is an unbalanced panel of 16 of the largest US banks over 2002–2013. These banks are large enough to have significant uninsured and insured deposits, active CDS markets, and enough branch/rate information to estimate deposit demand and calibrate supply.

Interpretation: The focus on large banks is intentional. Large banks account for a substantial share of deposits, are central to systemic fragility debates, and have observable CDS spreads that proxy for default beliefs.

3.2 Deposit quantities

- Deposit levels come from the FDIC's Statistics on Depository Institutions.
- The data distinguish insured and uninsured deposits.
- Observations are bank-quarter observations.
- In Table 1, average insured deposits equal about $141 billion, while average uninsured deposits equal about $160.8 billion.

Interpretation: The insured/uninsured split is central. Insured deposits should be insensitive to default risk, while uninsured deposits are potentially impaired in default and therefore run-prone.

3.3 Financial distress measure

- Bank distress is measured using five-year credit default swap spreads from Markit.
- CDS spreads are converted into risk-neutral default probabilities under a constant-hazard assumption.
- The average CDS spread in the sample is about 0.83 percent, with a standard deviation of 0.88 percent.

Interpretation: CDS spreads are market-based and daily, and they directly capture the market price of bank default protection. They are therefore better suited than balance-sheet ratios for measuring depositors' default beliefs.

3.4 Deposit rates

- Deposit-rate data come from RateWatch.
- The paper uses one-year certificates of deposit (CDs), measured across bank branches.
- CDs with minimum deposits of $10,000 proxy for insured deposit products, while CDs with minimum deposits of $100,000 proxy for products more likely to include uninsured deposits.
- Deposit spreads are CD rates minus the corresponding one-year Treasury rate.

Interpretation: The product distinction is not perfect, but it provides a way to observe different rates for accounts with different insurance exposure.

3.5 Bank service quality and differentiation controls

- Bank differentiation is controlled using branch counts from the FDIC and manually collected ATM counts.
- Service quality is also measured using consumer complaints per account from the CFPB Consumer Complaint Database.
- Bank fixed effects capture persistent differences in bank quality, brand, service networks, and depositor relationships.

Interpretation: This is why the model departs from homogeneous-bank run models. Depositors value bank services, so even a distressed banking system need not collapse completely: relatively safer banks can attract depositors.

4 Model and empirical specifications

4.1 Depositor utility

Depositors decide between banks. Insured depositors value rates and bank services, while uninsured depositors additionally dislike default risk:

$$
\begin{aligned}
u^I_{jkt} &= \alpha^I i^I_{kt} + \delta^I_k + \varepsilon^I_{jkt}, \\
u^N_{jkt} &= \alpha^N i^N_{kt} - \gamma\rho_{kt} + \delta^N_k + \varepsilon^N_{jkt}.
\end{aligned}
$$

Here i^I_{kt} and i^N_{kt} are insured and uninsured deposit rates, ρ_{kt} is the expected default probability, δ_k is bank quality, and ε is type-I extreme value.

Interpretation: The key demand parameter is γ. A larger γ means uninsured depositors are more run-prone because their demand falls more sharply when expected default probability rises.

4.2 Logit demand for deposits

The logit assumptions imply:

$$
\begin{aligned}
s^I_{kt} &= \frac{\exp(\alpha^I i^I_{kt} + \delta^I_k)}{\sum_{\ell=1}^K \exp(\alpha^I i^I_{\ell t} + \delta^I_\ell)}, \\
s^N_{kt} &= \frac{\exp(\alpha^N i^N_{kt} - \gamma\rho_{kt} + \delta^N_k)}{\sum_{\ell=1}^K \exp(\alpha^N i^N_{\ell t} - \gamma\rho_{\ell t} + \delta^N_\ell)}.
\end{aligned}
$$

Interpretation: The model treats insured and uninsured depositors as different markets. Insured deposit demand responds to rates and bank quality but not default. Uninsured deposit demand responds to rates, quality, and default probability.

4.3 Bank profits and default decision

Bank k earns stochastic returns $R_{kt} \sim N(\mu_k, \sigma_k)$ on deposits. It also incurs an additional cost c_k for insured deposits. Period profit before debt obligations is:

$$
\pi_{kt} = M^I s^I_{kt}(R_{kt} - c_k - i^I_{kt}) + M^N s^N_{kt}(R_{kt} - i^N_{kt}).
$$

Banks are also financed with long-term debt that requires a coupon payment. Equity holders inject funds or default depending on whether continuation value exceeds the injection cost.

Interpretation: Default is endogenous. A bank defaults not mechanically when returns are low, but when equity holders prefer default to recapitalization.

4.4 Equilibrium concept

The model studies stationary pure-strategy Bayesian Nash equilibria. Depositors form rational beliefs about default probabilities, choose deposit products, and banks set interest rates and default policies optimally. In equilibrium, depositors' beliefs about default match banks' default probabilities.

Interpretation: Multiple equilibria arise because beliefs about default affect uninsured deposit demand, which affects profitability, which affects actual default probability.

4.5 Reduced-form demand evidence

The paper first estimates regressions of log deposit shares on default probability:

$$\log s_{kt}^N = \gamma\rho_{kt} + \mu_k + \mu_t + \Gamma X_{kt} + \varepsilon_{kt}.$$

For the insured/uninsured difference-in-differences test, the paper compares responses of uninsured and insured deposit shares to the same bank distress shock.

Interpretation: This test is designed to address the concern that bank quality shocks, not default risk, drive deposit outflows. Insured deposits provide a placebo because they should not respond to default risk.

4.6 Supply-side calibration

Demand estimates plus banks' first-order conditions identify the supply-side parameters μ_k, σ_k, and c_k for each bank. The paper shows that, conditional on an observed equilibrium, the parameters are exactly and uniquely identified.

Interpretation: This is the bridge from demand estimation to policy analysis. Without supply-side calibration, one could estimate depositor run-proneness but not determine whether multiple equilibria are quantitatively possible.

5 Identification strategy

5.1 Identifying run-prone demand

The main empirical challenge is reverse causality: a drop in deposits lowers bank profitability and increases default risk, making it hard to infer whether distress causes withdrawals or withdrawals cause distress.

The paper addresses this in three ways:

- **Insured-deposit placebo:** insured deposits should not respond to default probability. Table 2 shows the uninsured response is large while insured response is small.
- **Fixed effects:** bank fixed effects absorb persistent service quality and brand; quarter fixed effects absorb aggregate deposit and crisis conditions.

- **Instrumental variables:** Table 3 instruments default probability using bank portfolio exposure to mortgage-related securities and loans, exploiting within-bank variation in portfolio composition and returns.

Interpretation: The identification goal is not to estimate a randomized treatment effect of bank distress. It is to isolate a structural demand elasticity needed to discipline the bank-run model.

5.2 Identifying supply-side parameters

The calibration uses revealed preference: observed interest rates, market shares, demand slopes, and default probabilities must satisfy banks' optimality conditions. The key maintained assumption is that the observed equilibrium is generated by banks maximizing equity value in the model.

Interpretation: Supply identification is model-based. It depends on the assumption that banks' first-order conditions and default choices are well captured by the static stationary model.

5.3 Identifying multiple equilibria

After estimating demand and calibrating supply, the authors solve for all equilibrium belief/default/rate configurations supported by the same fundamentals. This is not an empirical treatment design but an equilibrium computation.

Interpretation: The multiple-equilibrium analysis asks: holding fundamentals fixed, are there other self-consistent beliefs and bank choices that would generate much worse outcomes? The answer is yes.

5.4 Identifying policy effects

Policy experiments are counterfactual simulations. The authors change FDIC insurance, rate caps, capital requirements, too-big-to-fail expectations, adjustment costs, or asset composition, and recompute equilibria.

Interpretation: The policy results are model-based. Their credibility depends on how well demand estimates and supply calibration capture behavior outside the observed equilibrium.

6 Tables and figures

6.1 Figure 1: Deposits versus Financial Distress: Citibank and JPMorgan Chase

Read:

- Panel A shows that as Citibank distress rises relative to JPMorgan, Citi's uninsured deposit share falls and JPMorgan's uninsured share rises.
- Panel B shows no comparable pattern for insured deposits.
- The figure motivates the central demand restriction: uninsured deposits are run-prone; insured deposits are not.

Interpretation: Figure 1 is not the final structural estimate, but it gives the raw intuition. Default risk matters for uninsured deposits because depositors may lose money or face disruption in default; it does not matter much for insured deposits because the FDIC absorbs that risk.

Economic message: Depositor runs are not just theoretical. Market shares of uninsured deposits move away from distressed banks in the data.

Panel A. Uninsured deposits

Panel B. Insured deposits

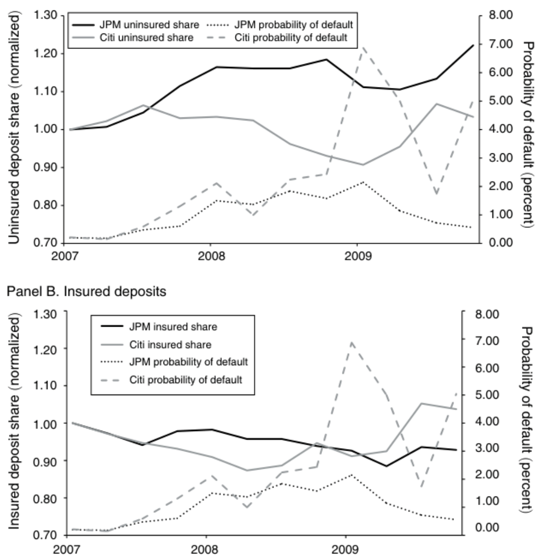

FIGURE 1. DEPOSITS VERSUS FINANCIAL DISTRESS: CITIBANK AND JPMORGAN CHASE

TABLE 1—DEPOSIT LEVEL, INTEREST RATE, AND CDS SUMMARY STATISTICS

Variable	Observations	Mean	Standard deviation	Min.	Max.
Insured deposits ($bn)	566	141.0	162.0	11.27	845.6
Uninsured deposits ($bn)	566	160.8	205.2	4.083	939.0
CDS spread	566	0.83%	0.88%	0.05%	5.47%
Deposit spread (min. dep. = $10k)	566	−0.31%	0.71%	−2.66%	2.03%
Deposit spread (min. dep. = $100k)	564	−0.22%	0.70%	−3.67%	2.03%

Notes: Table 1 displays the summary statics corresponding to our retail bank dataset, which consists of an unbalanced panel of 16 of the largest US banks over the period 2002–2013. Observations are bank by quarter. We measure the level of insured and uninsured deposits as per the FDIC's Statistics on Depository Institutions. We measure CDS at the monthly level using the average daily CDS spread for the five-year CDS contract as reported by Markit. We measure the deposit rate offered by each bank using the median one-year CD rate offered by each bank across all of its branches in a given month as reported by RateWatch. We separately examine the certificate of deposit rates offered for accounts with minimum deposit amounts of $10k and $100k. We report the deposit rate spread, which reflects the one-year certificate of deposit rate relative to the corresponding one-year Treasury rate.

6.2 Table 1: Deposit Level, Interest Rate, and CDS Summary Statistics

Read:

- The dataset has 566 bank-quarter observations.
- Average insured deposits are about $141 billion, while average uninsured deposits are about $160.8 billion.
- The average CDS spread is 0.83 percent, with a standard deviation of 0.88 percent.
- Deposit spreads are negative on average relative to the one-year Treasury, but $100k-minimum CDs have higher spreads than $10k-minimum CDs.

Interpretation: Table 1 shows that uninsured deposits are large enough to matter for bank funding and financial fragility. The table also shows that market-implied default risk varies substantially across banks and time.

Economic message: The paper studies a setting where run-prone funding is large, observable, and important for the largest banks.

6.3 Table 2: Deposits and Financial Distress

Read:

- The coefficient on default probability is −1.98 for the uninsured-minus-insured difference.
- The uninsured-deposit coefficient is −2.13.
- The insured-deposit placebo coefficient is only −0.16.
- The uninsured results include quarter fixed effects and bank fixed effects.

Interpretation: Table 2 establishes the key reduced-form fact. Distress reduces uninsured deposits, but it does not reduce insured deposits. This helps isolate financial distress from unobserved declines in bank service quality.

Economic message: Uninsured deposits discipline banks and create run risk; insured deposits are stabilized by government insurance.

Table 2—Deposits and Financial Distress

	Uninsured–insured deposits	Uninsured deposits	Insured deposits
Probability of default	−1.98	−2.13	−0.16
	(0.96)	(1.14)	(1.04)
Share difference (uninsured − insured)	X		
Quarter fixed effects	X	X	X
Bank fixed effects	X	X	X
Observations	566	566	566
R^2	0.949	0.970	0.948

Notes: Table 2 displays the regression of a bank's logged deposit share on its probability of default. Each regression is estimated using an unbalanced panel of 16 of the largest US banks with quarterly observations over the period 2002–2013. All specifications control for the number of bank branches. Robust standard errors in parentheses.

Table 3—Demand Estimates

	Demand estimates				Placebo for default prob		
	Unins. deposits			Ins. deposits	Ins. deposits		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Deposit rate	12.62	18.21	16.64	58.79	59.70	54.96	55.35
	(7.49)	(6.08)	(5.80)	(8.16)	(8.63)	(8.14)	(8.18)
Probability of default	−18.61	−10.66	−12.60		4.01	−2.86	−1.79
	(7.21)	(5.83)	(5.29)		(5.66)	(5.33)	(4.33)
IV-0 (pass-through)	X	X	X	X	X	X	X
IV-1 (CMOs)	X		X		X		X
IV-2 (loans)		X	X			X	X
Observations	564	564	564	566	566	566	566
R^2	0.958	0.966	0.964	0.917	0.915	0.921	0.921

Notes: Table 3 displays the demand estimates for uninsured and insured deposits. The dependent variable in each specification is the log of a bank's market share. Each demand specification is estimated using an unbalanced panel of 16 of the largest US banks with quarterly observations over the period 2002–2013. Each observation is weighted by the square root of the market size. All specifications include bank and quarter fixed effects and control for the number of bank branches. Robust standard errors in parentheses.

6.4 Table 3: Demand Estimates

Read:

- The demand estimates show that uninsured deposits respond positively to deposit rates and negatively to default probability.
- In the preferred uninsured specification, the default probability coefficient is approximately −12.60.
- The insured-deposit placebo default coefficients are small and insignificant.
- The deposit-rate coefficients imply that banks compete in deposit rates, and the default-risk coefficients imply that uninsured demand is run-prone.

Interpretation: Table 3 supplies the structural demand elasticity used in the model. It is the paper's most important estimation table because the possibility of multiple equilibria depends quantitatively on the size of the uninsured-deposit response to default.

Economic message: The run-prone elasticity is large enough to make a belief shock self-fulfilling in the calibrated banking model.

6.5 Figures 2–4: Calibrated Deposit Return Parameters

Read:

- Figure 2 displays the calibrated noninterest cost of insured deposits. Costs are around 4–6 percent across banks.
- Figure 3 displays the calibrated mean return on deposits. Mean returns are around 7.5–8.5 percent.
- Figure 4 displays calibrated standard deviations of deposit returns. The distribution ranges roughly from 8 percent to above 18 percent.

Interpretation: These figures evaluate whether the supply calibration is plausible. The calibrated return means and costs are not directly used as observed balance-sheet moments but are consistent with independent estimates of bank profitability and account-servicing costs.

Economic message: The structural model is disciplined by the observed behavior of banks and depositors, and the implied cost/return primitives are economically plausible.

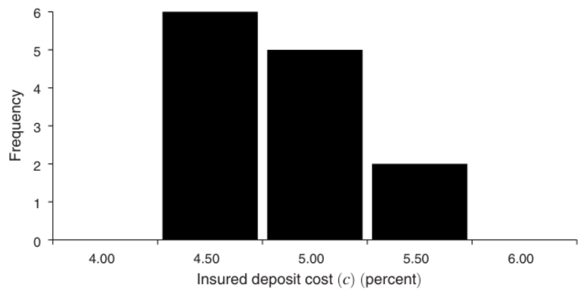

Figure 2. Calibrated Noninterest Cost of Insured Deposits (*As of March 31, 2008*)

(a) Figure 2: noninterest cost of insured deposits.

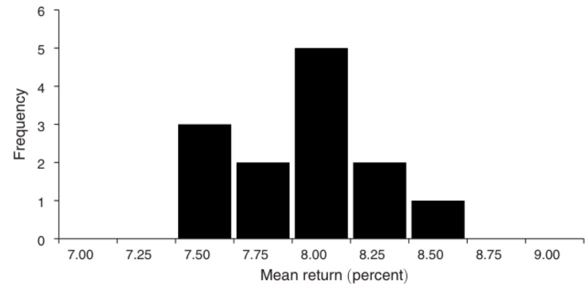

Figure 3. Calibrated Mean Return on Deposits (*As of March 31, 2008*)

(b) Figure 3: mean return on deposits.

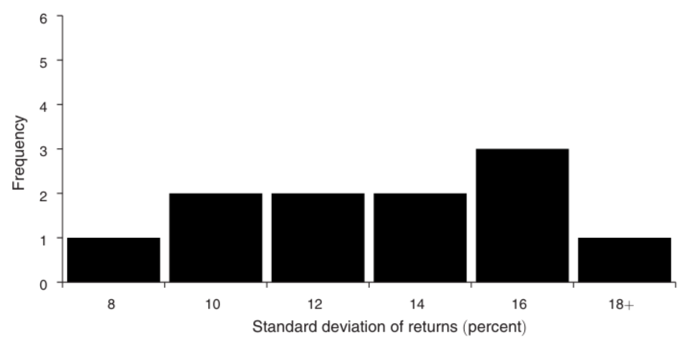

Figure 4. Calibrated Standard Deviation of Return on Deposits (*As of March 31, 2008*)

Figure 4: calibrated standard deviation of return on deposits.

6.6 Table 4: Multiple Equilibria 2008

Read:

- The observed March 2008 equilibrium has low default probabilities for the five banks shown.
- Alternative equilibria exist in which individual banks have default probabilities above 40 percent: Wells Fargo in equilibrium 3, Bank of America in equilibrium 4, Wachovia in equilibrium 5, JPMorgan in equilibrium 6, and Citibank in equilibrium 7.
- Bad equilibria raise expected FDIC insurance costs and reduce welfare by hundreds of billions to more than $1 trillion.
- In bad equilibria, distressed banks often set high insured deposit rates and attract insured deposits because the FDIC guarantee makes those deposits cheap from shareholders' perspective.

Interpretation: Table 4 is the central equilibrium result. The same fundamentals can support a stable observed equilibrium and much worse self-fulfilling equilibria. The mechanism is not direct balance-sheet contagion; it is deposit competition and belief-driven withdrawal of uninsured deposits.

Economic message: Banking stability is equilibrium dependent. A bank can become fragile because depositors believe it is fragile, and the resulting deposit outflow validates that belief.

TABLE 4—MULTIPLE EQUILIBRIA 2008 (*Percent*)

	Observed eq.	(2)	(3)	(4)	(5)	(6)	(7)
Insured interest rate							
JPMorgan Chase	1.73	0.98	2.46	2.65	2.44	10.48	3.17
Bank of America	1.98	1.53	2.13	7.33	2.14	2.44	2.46
Wells Fargo	2.13	2.05	10.04	3.06	3.04	3.58	3.68
Citibank	2.23	2.11	3.01	3.21	2.98	3.72	12.26
Wachovia	2.08	2.04	2.59	2.62	8.75	2.93	2.98
Uninsured interest rate							
JPMorgan Chase	1.73	0.94	2.41	2.56	2.39	20.35	3.02
Bank of America	1.97	1.40	1.94	11.43	1.96	2.23	2.24
Wells Fargo	2.32	2.25	17.41	3.21	3.20	3.71	3.81
Citibank	2.23	2.13	2.94	3.09	2.92	3.52	24.35
Wachovia	2.23	2.19	2.67	2.71	14.06	3.00	3.04
Probability of default							
JPMorgan Chase	1.50	0.19	2.86	3.29	2.83	**48.35**	4.37
Bank of America	1.82	0.03	1.85	**53.34**	1.94	3.21	3.27
Wells Fargo	1.50	1.34	**46.61**	3.56	3.49	4.81	5.06
Citibank	2.11	1.92	3.36	3.74	3.33	4.63	**48.20**
Wachovia	3.28	3.14	4.74	4.92	**52.08**	5.96	6.12
Insured deposit share							
JPMorgan Chase	3.38	2.26	1.00	1.84	1.30	83.89	0.91
Bank of America	9.26	7.39	1.94	68.34	2.57	1.75	1.42
Wells Fargo	3.99	3.96	80.41	2.19	1.73	1.35	1.14
Citibank	2.07	2.00	0.63	1.17	0.82	0.72	86.68
Wachovia	5.81	5.89	1.51	2.53	74.47	1.38	1.14
Cumulative share	24.51	21.50	85.49	76.08	80.88	89.09	91.29
Uninsured deposit share							
JPMorgan Chase	15.86	16.06	15.96	16.62	16.06	1.19	17.09
Bank of America	9.23	10.32	9.76	0.08	9.75	10.03	10.04
Wells Fargo	4.30	4.25	0.19	4.40	4.16	4.43	4.39
Citibank	16.80	16.58	17.23	18.04	17.33	18.79	2.51
Wachovia	4.74	4.70	4.52	4.77	0.08	4.76	4.73
Cumulative share	50.93	51.91	47.67	43.91	47.36	39.19	38.77
FDIC insurance cost	$14bn	$8bn	$1,002bn	$979bn	$1,037bn	$1,086bn	$1,118bn
Change in welfare	—	$20bn	−$1,143bn	−$1,205bn	−$1,221bn	−$1,332bn	−$1,365bn

Notes: The first column displays the observed equilibrium as of March 31, 2008. Column 2 displays the best potential equilibrium in terms of welfare. Columns 3–7 display potential bank-run equilibria. Equilibria are ordered by welfare. As detailed in the online Appendix, welfare is reported relative to its respective values in the observed equilibrium, assuming a bankruptcy cost of 20 percent. We do not report all potential equilibria. For ease of exposition, we omit equilibria where the cumulative difference in default probabilities is within 10 percent of a reported equi-

TABLE 5—COUNTERFACTUAL ANALYSIS: FDIC INSURANCE (*Percent*)

Bank	Probability of default	Counterfactual	Δ Insurance cost
Panel A			
JPMorgan Chase	1.50	1.51	$50m
Citibank	2.11	2.13	$76m
Bank of America	1.82	1.84	$56m
Wells Fargo	1.50	1.51	$17m
Wachovia	3.28	3.30	$67m
Panel B			
JPMorgan Chase	1.50	1.37	−$95m
Citibank	2.11	2.02	−$51m
Bank of America	1.82	1.72	−$174m
Wells Fargo	1.50	1.42	−$67m
Wachovia	3.28	3.20	−$82m

Notes: Panel A: Column 1 displays the realized equilibrium probability of default as of March 31, 2008. Column 2 displays an equilibrium probability of default if the FDIC were to insure an additional 1 percent of uninsured deposits. The additional insured deposits are assumed to be treated as a new type of deposit that is valued by consumers similarly to uninsured deposits, except that depositors are insensitive to default risk (i.e., $\gamma = 0$). We calculate and select the reported new equilibrium using a nonlinear equation solver (we use the R package NLEQSLV with Broyden's Method) initiated at the observed equilibrium. Column 3 displays the change in the hypothetical equilibrium cost of the FDIC policy change relative to the old policy. We calculate the cost change as the difference in expected insurance payout. We calculate the expected insurance payout as the weighted sum of insured deposits weighted by the probability of default, assuming a 40 percent recovery rate. Negative values represent a surplus to the FDIC. Panel B: Column 1 displays the realized equilibrium probability of default as of March 31, 2008. Column 2 displays an equilibrium probability of default if the FDIC were to insure an additional 1 percent of uninsured deposits. The additional insured deposits are assumed to be treated identically to existing insured deposits. We calculate and select the reported new equilibrium using a nonlinear equation solver (we use the R package NLEQSLV with Broyden's Method) initiated at the observed equilibrium. Column 3 displays the change in the hypothetical equilibrium cost of the FDIC policy change relative to the old policy. We calculate the cost change as the difference in expected insurance payout. We calculate the expected insurance payout as the weighted sum of insured deposits weighted by the probability of default, assuming a 40 percent recovery rate. Negative values represent a surplus to the FDIC.

6.7 Table 5: Counterfactual Analysis: FDIC Insurance

Read:

- Panel A considers insuring an additional 1 percent of uninsured deposits while treating them as a new product valued like uninsured deposits except for insurance.
- Panel B considers insuring an additional 1 percent of uninsured deposits while treating them as identical to existing insured deposits.
- Depending on how newly insured depositors are modeled, the policy can increase costs or create a surplus for the FDIC.

Interpretation: The table shows that deposit-insurance expansion is not a mechanically stabilizing policy. Its effects depend on what type of depositor is being newly insured and whether those depositors value banking services like insured or uninsured depositors.

Economic message: Deposit insurance policy has distributional consequences and depends critically on depositor composition.

6.8 Figure 5: Capital Requirements

Read:

- Panel A shows mean default probabilities in the best and worst equilibria as capital requirements vary.
- Panel B shows welfare: worst-equilibrium welfare improves sharply as capital requirements rise toward roughly 18 percent, then improves more slowly and eventually declines.

- Panels C–E decompose welfare into consumer surplus, flow to equity, and expected FDIC losses.
- The paper reports that the max-min optimal capital requirement in the baseline is about 39 percent, but the robust recommendation is closer to 15–18 percent across perturbations.

Interpretation: Figure 5 shows why policy evaluation in a multiple-equilibrium model differs from standard comparative statics. A policy can eliminate disastrous equilibria but also reduce consumer surplus and valuable bank intermediation.

Economic message: Banking stability and welfare are not identical. Regulators should care about both default probabilities and the services/value generated by the banking system.

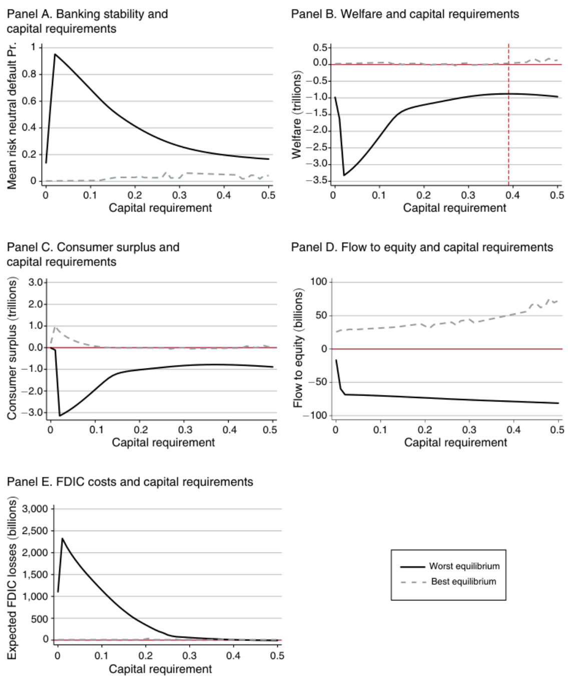

FIGURE 5. CAPITAL REQUIREMENTS

Notes: Figure plots the probability of default (averaged across banks), welfare, consumer surplus, annualized equity value, and expected FDIC losses in the worst and best equilibria as of March 31, 2008. The best/worst equilibria are defined in terms of the relevant measure (probability of default, welfare, etc.). As detailed in the online Appendix.

TABLE 6

Panel A. Alternative specifications, calibrated parameters	Mean return (μ)	Standard deviation (σ)	Noninterest cost (c)
Baseline model	7.80%	15.94%	4.67%
Alternative specifications			
Capital req.(8%)	7.56%	18.38%	4.67%
Capital adj. costs (5%)	8.19%	11.35%	4.67%
TBTF (50%)	7.40%	14.72%	4.67%
Capital rq., adj. and TBTF (50%)	7.68%	13.38%	4.67%

Panel B. Alternative specifications, optimal capital requirement	Optimal capital req. κ
Baseline model	39%
Alternative specifications	
Insured depositor run	22%
Capital req.(8%)	42%
Capital adj. costs (5%)	16%
TBTF (50%)	24%
Capital rq., adj. and TBTF (50%)	24%
Capital rq., adj., TBTF (50%) and bankruptcy costs (%)	24%

Notes: Panels A and B display the calibrated parameters and optimal capital requirements under the alternative model specifications as of March 31, 2008. Panel A displays the average of the calibrated parameters (μ, σ, c) under each specification. Panel B displays the optimal capital requirement. The optimal capital requirement maximizes welfare, given that the worst equilibrium outcome (in terms of welfare) is realized. The alternative model specifications reported in panel A are as follows. First, we calibrate the model to existing capital requirements of 8 percent. Second, we calibrate the model where investors anticipate a "too big to fail" (TBTF) policy where the government bails out uninsured depositors with 50 percent probability. Third, we calibrate the model with capital adjustment costs (deadweight cost of external financing), which is proportional to the amount of funds injected, with a constant marginal cost of 5 percent. And last, we calibrate the model to existing capital requirements, under the TBTF policy and with capital adjustment costs. We examine the optimal capital requirements if insured depositors are run prone (sensitivity $\gamma^I = 0.5\gamma$) and with bankruptcy costs of 20 percent. The addition of bankruptcy costs and/or run-prone insured depositors does not impact the calibration of the model. The details of each alternative specification are discussed in Section VII and the online Appendix.

6.9 Table 6: Alternative Specifications and Optimal Capital Requirements

Read:

- Panel A reports average calibrated parameters across model extensions: baseline, 8 percent existing capital requirements, capital adjustment costs, too-big-to-fail expectations, and combinations.
- Panel B reports the optimal capital requirement under each model variation.
- The baseline max-min optimum is 39 percent, but several richer specifications have optimal requirements around 16–24 percent.

Interpretation: Table 6 qualifies the baseline policy recommendation. The exact 39 percent level is not robust, but the result that very low capital requirements allow severe bad equilibria is robust.

Economic message: A prudent policy lesson is that requirements below roughly 15–18 percent can expose the system to large equilibrium-selection losses.

6.10 Figure 6: Capital Requirements under Alternative Specifications

Read:

- Figure 6 plots best- and worst-equilibrium welfare under multiple model perturbations.
- Panel A includes an 8 percent existing capital requirement; Panel B includes too-big-to-fail beliefs; Panel C includes capital adjustment costs; Panels D and E combine features; Panel F allows insured depositors to become run-prone.
- Across perturbations, welfare in the worst equilibrium is very low at small capital requirements and improves as requirements increase.

Interpretation: The figure shows robustness of the lower-bound policy insight. Exact optima vary, but the model repeatedly warns against low capital requirements when equilibrium selection is uncertain.

Economic message: Robust policy should be based less on a point estimate of the optimal requirement and more on avoiding regions where bad equilibria impose very large welfare losses.

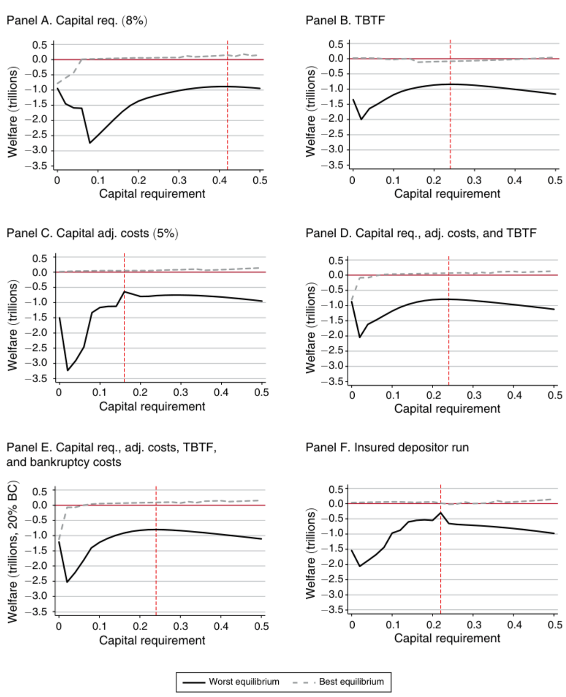

Figure 6. Capital Requirements Under Alternative Specifications

Notes: Panels A–F plot the welfare in the worst and best equilibria as of March 31, 2008 as a function of capital requirements for each model perturbation. The best/worst equilibria are defined in terms of welfare. As detailed in the online Appendix, welfare, consumer surplus, annualized equity value, and expected FDIC losses are reported relative to their respective values in the observed equilibrium. The model perturbations reported in panels A–F are as follows. In panels A, D, and E, we calibrate to existing capital requirements of 8 percent. In panels B, D, and E, we allow for investors to anticipate a TBTF policy where the government bails out uninsured depositors with 50 per-

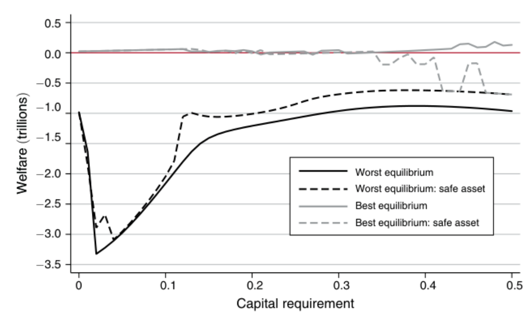

Figure 7. Risk-Free Capital Requirements

lotes: Figure plots the welfare in the worst and best equilibria as of March 31, 2008 as a function of capital requi
nents. The best/worst equilibria are defined in terms of welfare. The black and gray solid lines plot welfare in t

6.11 Figure 7: Risk-Free Capital Requirements

Read:

- Figure 7 compares baseline capital requirements, where required capital is co-invested in the risky bank asset, with capital invested in a safe asset.
- Safe-asset requirements bound the welfare outcomes between best and worst equilibria more tightly.
- There is still a large drop in worst-equilibrium welfare below roughly 14 percent.

Interpretation: The implementation of capital requirements matters. Capital requirements are not only a number; they also depend on what assets capital can be invested in and how those assets affect bank risk and returns.

Economic message: Regulatory design details can change the trade-off between banking services, shareholder value, default risk, and FDIC costs.

7 Interpretive synthesis

- **Demand-side discipline.** Uninsured depositors respond to distress, so they can run. Insured depositors do not respond to distress, so they stabilize funding but create moral-hazard-like rate competition.
- **Supply-side amplification.** Distressed banks optimally raise rates because marginal deposits are valuable when equity holders expect to default with high probability. Competitors respond by raising rates, lowering margins system-wide.
- **Equilibrium selection.** The same fundamentals can support stable and unstable equilibria. Policy analysis must therefore evaluate the set of equilibria, not just the observed one.
- **Product differentiation as stabilizer.** Differentiated bank services prevent every bank from becoming unstable at once; depositors move to banks whose services and safety remain attractive.
- **Regulatory implication.** Capital requirements, rate caps, and deposit insurance interact with competitive forces. Policies can reduce default risk while lowering welfare or changing service provision.

8 Short summary

- The paper estimates demand for insured and uninsured deposits at large US banks.
- Uninsured deposits decline with bank distress; insured deposits do not.
- The estimated run-prone elasticity is embedded in a structural model of differentiated deposit competition and endogenous default.
- The calibrated model has multiple equilibria with bank-run features.
- In March 2008, Wachovia's observed default probability was about 3.3 percent, while another equilibrium supported about 52 percent default probability.
- Bad equilibria can spread distress through insured-deposit rate competition, even without direct bank balance-sheet linkages.
- The system does not completely collapse in the model because depositors value banking services and shift toward safer banks.
- FDIC expansion, rate caps, and capital requirements have nontrivial and sometimes counterintuitive effects.
- Capital requirements below roughly 15–18 percent leave the model vulnerable to bad equilibria with large welfare losses.
- The paper's incremental contribution is a quantitative, estimated, policy-usable bank-run model built on depositor demand and bank competition.

9 Conclusion

9.1 Core conclusion

Egan, Hortaçsu, and Matvos show that uninsured-deposit run risk can be quantitatively important in the US banking sector. The estimated demand elasticity is large enough that the calibrated banking system supports multiple equilibria, including bad equilibria with high default probabilities and large welfare losses.

9.2 Economic intuition

A belief that a bank is risky reduces demand for its uninsured deposits. The deposit outflow lowers profitability and increases the bank's actual default probability. Insured deposits create a different distortion: distressed banks can offer high insured deposit rates because the FDIC absorbs downside risk, forcing other banks to compete and potentially spreading distress.

9.3 Incremental contribution in one sentence

The paper's distinctive contribution is to connect empirical deposit demand, differentiated-bank competition, endogenous default, multiple-equilibrium run theory, and counterfactual banking regulation in one structural model.

9.4 Final takeaway

The paper changes how one should evaluate bank regulation. A policy that looks good in the observed equilibrium may behave very differently in a bad equilibrium, and policies that increase stability may lower welfare by reducing valuable banking services. Quantitative bank-run policy analysis therefore requires both demand estimation and equilibrium computation.